

CSE420 Final Fall '23

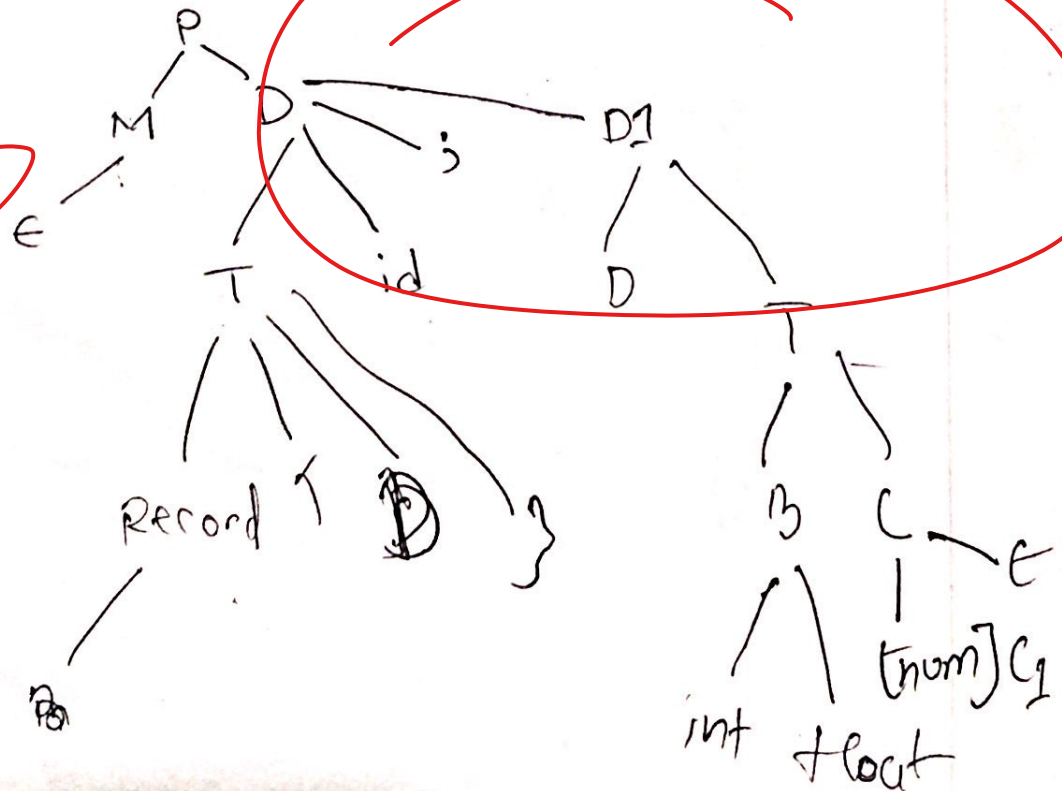
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Date: 18/12/23

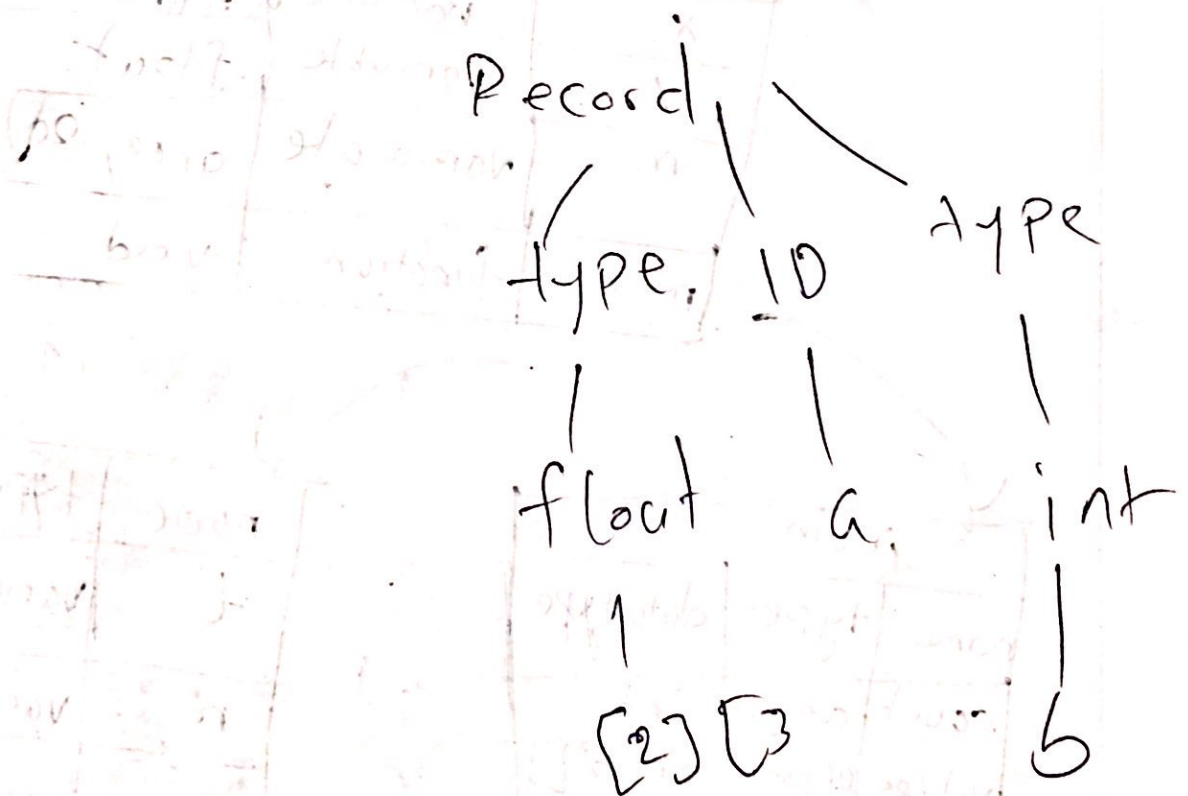
Answer to question - 1



Record: { float (2) (3) a; int b; }

Offset = float (2) (3)

= 2 x 3 x 4 = 24 bytes.



Answer to question - 2

Scope management:

Global Scopes		
name	type	data type
x	variable	float
v	variable	float
m	variable	array (2d)
main	function	void

main		
name	type	data type
count	argument	int.
values	argument	char*[]

name	type	data type
t	variable	float
n	variable	int
s	variable	float
c	variable	int
i	variable	int
j	variable	int
v	variable	int
count	variable	int
m	variable	float

name	type	data type
i	variable	int.
s	variable	float

from(j) for

Name	type	datatype
j	variable	int
v	variable	int
count	variable	int
n	variable	float

from(i) loop

Name	type	datatype
i	variable	int
s	variable	float
c	variable	int

while

Name	type	data type
m	variable	float

Since already declared variables are present

Answer to question - 3

(a) ~~Three~~³ address code :

$$\begin{aligned} t1 &= a + a \\ t2 &= a + a \\ t3 &= t1 - t2 \\ a &= t3 + a \end{aligned}$$

$$t1 = a + a$$

$$t2 = a + a$$

$$t3 = t1 - t2$$

$$a = t3 + a$$

Quadruple representation,

Sl	Operator	Arg 1	Arg 2	result
①	+	a	a	t1
②	+	a	a	t2
③	-	t1	t2	t3
④	+	t3	a	a

Triple :

Sl	Operator	Arg 1	Arg 2
0	+	a	a
1	+	a	a
2	-	(0)	(1)
3	+	(2)	a